

Nikhilesh Vijay Chirutala

Backend Software Engineer

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SUMMARY

Backend Software Engineer with 2.8+ years of experience building and maintaining production applications across AI, CRM, trading, and cloud contact center platforms. Strong experience with Python, FastAPI, Django, Flask, REST API development, MongoDB, PostgreSQL, MySQL, AWS, and backend application development. Experienced in designing APIs, implementing business logic, integrating databases and third-party services, building data-processing workflows, authentication and authorization, cloud storage workflows, observability, and production troubleshooting. Hands-on experience developing RAG-based chatbot solutions and integrating LLM capabilities into backend applications. Proven ability to develop reliable backend services, migrate legacy workflows, troubleshoot production issues, and support deployments across distributed environments.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (ES6+), TypeScript

Backend & Frameworks: FastAPI, Django, Django REST Framework, Flask, REST APIs, API Development, Backend Automation, Business Logic

Databases: MongoDB, PostgreSQL, MySQL, Neon DB, SQL, CRUD Operations, Data Modeling

Cloud & Infrastructure: AWS, Amazon S3, Linux, Vercel, GitHub Actions, CI/CD

APIs & Integrations: REST APIs, Twilio, GitHub API, API Integration, Postman

Security & Authentication: JWT, Authentication, Authorization, RBAC, ACL

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering

Observability & Production: Datadog, Sentry, Monitoring, Debugging, Production Troubleshooting, Error Handling

Testing & Development Tools: Postman, Unit Testing, Jest, Git, GitHub, Debugging

Frontend: React.js, Next.js, JavaScript, TypeScript, Jinja2, Tailwind CSS, Bootstrap

WORK EXPERIENCE

Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, MySQL, Datadog

- Develop and maintain production-grade Python services supporting call-recording ETL, data processing, storage workflows, and backend operations for a high-volume cloud contact center platform.
- Migrated a legacy Perl-based call-recording workflow to a long-running Python service deployed across 40 production clusters, implementing feature flags, staged rollout, validation, rollback, and monitoring.
- Built Python service integrations with a Call Intelligence microservice, implementing APIs, eligibility checks, account and campaign controls, daily processing limits, retries, response validation, and result persistence.
- Developed backend validation and recovery workflows for call-recording data, handling incomplete processing, failed operations, and inconsistent storage states across production environments.
- Built AWS S3 audit and recovery workflows for recording storage, implementing deterministic object handling and idempotent re-upload operations for incomplete or failed storage migrations.
- Supported production deployment and rollout of Python services across multiple clusters, including feature-flag configuration, validation, monitoring, and troubleshooting during migration activities.
- Implemented production observability using Datadog, and Sentry to monitor service health, track application errors, diagnose processing failures, and support incident resolution.
- Investigated production issues across Python services, APIs, databases, and cloud storage, performing root-cause analysis, implementing fixes, and improving reliability of backend workflows.

Software Engineer • Mantra Technologies

Feb 2024 – Mar 2026 | Hyderabad, India

- Developed backend services, REST APIs, business logic, and application workflows using Python, FastAPI, and MongoDB across multiple production applications.
- Designed and implemented API-driven backend modules for application-specific workflows, data processing, authentication, authorization, and third-party integrations.
- Implemented authentication, RBAC, and ACL-based authorization to enforce resource-level access controls across application services.
- Integrated third-party services including Twilio and GitHub API, handling API communication and application-level integration workflows.
- Worked with MongoDB for application data operations and developed backend workflows involving data creation, updates, retrieval, validation, and persistence.
- Maintained GitHub Actions CI/CD workflows and supported application deployments across AWS and Vercel environments.
- Performed API testing, debugging, defect resolution, and production support to maintain application reliability.

Key Projects:

AI Bull — AI-Powered Trading Platform

FastAPI, Python, MongoDB, RAG, LLM, React.js, Next.js

- Developed Python and FastAPI backend services for an AI-powered trading and investment analysis platform supporting stock and mutual fund analysis, screening, and back-testing.
- Implemented backend business logic for financial data processing, configurable screening, filtering, and application workflows.
- Developed a Retrieval-Augmented Generation (RAG) based chatbot that retrieves relevant financial and application-specific information to generate contextual responses.
- Integrated RAG capabilities into application workflows, enabling conversational access to relevant financial information through backend services.
- Built back-testing engines and configurable processing workflows for stock and mutual fund analysis.
- Integrated MongoDB for application data persistence and supported API-driven communication between backend services and application interfaces.

CRM.io — AI-Driven CRM System

FastAPI, Python, MongoDB, ACL, Twilio, REST APIs, Jinja2

- Developed backend CRM modules using FastAPI and Python for Contacts, Tasks, communication workflows, and telephony operations.
- Designed and implemented REST APIs and business logic for CRM workflows and application resources.
- Implemented RBAC and ACL authorization to control access to CRM resources based on user permissions.
- Integrated Twilio for telephony and communication functionality, handling backend communication workflows.
- Implemented MongoDB-backed data operations for CRM entities and application workflows.
- Tested APIs and backend workflows using Postman and resolved application defects across CRM and production workflows.

GharPe — Real Estate Management Platform

FastAPI, Python, MongoDB, REST APIs, React.js

- Developed FastAPI/Python REST APIs supporting real estate project management, configuration, administrative workflows, and data operations.
- Implemented backend workflows for creating, updating, configuring, and managing real estate projects displayed across the platform.
- Developed lead management functionality for tracking and managing customer inquiries generated through the platform.
- Built backend functionality for builder management and association of builders with their respective projects.
- Integrated MongoDB for application data persistence and supported API-driven communication with the administrative application.

PERSONAL PROJECTS

Enterprise REST API Platform — Django and Flask Backend Application

Python, Django, Django REST Framework, Flask, PostgreSQL, REST APIs

- Built modular REST APIs using Django, Django REST Framework, and Flask for authentication, business logic, and data-driven workflows.
- Integrated PostgreSQL with database models, relationships, CRUD operations, validation, and API error handling.
- Implemented JWT authentication and role-based access control for protected API resources.
- Tested and validated REST APIs using Postman, covering API behavior, authentication, request validation, and error handling.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform supporting projects, tasks, and team workflows.
- Implemented authentication and user management using Clerk and persistent application data using Neon DB.
- Developed application workflows for managing project and task data with persistent database storage.

JS Studio — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application using Next.js and Vercel.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur